Task 2: Focusing through a simulated bench

PhotonicsAI, Homework 2. Code and video:
https://www.loom.com/share/b2de470ccdab4941ae24e55aeceea0f9

https://github.com/LucyKhadrlian/Photonic-AI-HW-2_Task-2

Setup

The bench is simulated in Python. The DMD has 64x64 binary (0/1) amplitude pixels, the scattering medium is a random complex Gaussian transmission matrix T, and the camera has 32x32 pixels. A shot is E_out = |T E_in|^2 with 5% multiplicative noise, so repeated shots of the same pattern differ. The search follows the homework algorithm: group the DMD into macropixels, start from a random pattern, flip some macropixels, measure, and keep the flip if the enhancement rose. The focus is a disk at the camera centre and the background is a ring split into sectors. The enhancement is the mean focus intensity divided by each sector's mean intensity, averaged over the sectors. The reported value is the mean of 5 fresh shots after the search. The tool run_focusing is called by the executor of the Homework 1 agent (orchestrator, scientist, general, executor, running on a Groq model). The agent chose every setting itself.

Experiments run by the agent

Run	Macropixel	Focus radius	Background radius	Regions	Iterations	Flips per step	Enhancement
1	1	2	8	8	30	10	1.0
2	2	3	10	12	20	10	1.2
3	4	4	12	16	15	10	0.9
4	8	5	14	20	30	10	1.4

Comparison with the professor's result (about 33)

The best experimental enhancement is about 33. The best agent run reached 1.4, about 4% of that. An enhancement near 1 means no focus: the spot is as bright as the surrounding speckle. The cause is visible in the table. The agent chose only 15 to 30 iterations, while the search needs thousands, since each iteration tests one noisy shot. It also flipped 10 macropixels per step, which makes accepted improvements rare, and its large focus radii (up to 5 pixels) spread the light over many speckle grains.

As a manual reference, not an agent run, I ran the same tool with macropixel 4, focus radius 1, background radius 10, 8 regions, 20000 iterations and 1 flip per step. It gave an enhancement of 13.9, about 42% of the experimental value, with a clear bright spot at the centre. Earlier manual tests with fewer iterations gave about 6.5 and 11.9, so the enhancement grows with iterations and with a smaller focus region.

The remaining gap to 33 is probably due to: the limited number of controlled macropixels, since the enhancement grows roughly with it; binary amplitude modulation, which wastes part of the light

```
[experiments this turn]
- {'run': 1, 'macropixel': 1, 'focus_r': 2, 'bg_r': 8, 'regions': 8, 'iters': 30, 'flips': 10, 'enhanceme
nt': 1.0, 'image': 'focus_work\\focus_run_1.png'}
- {'run': 2, 'macropixel': 2, 'focus_r': 3, 'bg_r': 10, 'regions': 12, 'iters': 20, 'flips': 10, 'enhance
ment': 1.2, 'image': 'focus_work\\focus_run_2.png'}
- {'run': 3, 'macropixel': 4, 'focus_r': 4, 'bg_r': 12, 'regions': 16, 'iters': 15, 'flips': 10, 'enhance
ment': 0.9, 'image': 'focus_work\\focus_run_3.png'}
- {'run': 4, 'macropixel': 8, 'focus_r': 5, 'bg_r': 14, 'regions': 20, 'iters': 30, 'flips': 10, 'enhance
ment': 1.4, 'image': 'focus_work\\focus_run_4.png'}
------------------------------------------------------------
[Orchestrator] decision 3: next=finish | instruction=Stop the session as we have completed the required e
xperiments and obtained enhancement results. | reason=At least three experiments have been run and the en
hancement data is available, fulfilling the task.
------------------------------------------------------------

| run | macropixel | focus_r | bg_r | regions | iters | flips | enhancement |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 8 | 8 | 30 | 10 | 1.0 |
| 2 | 2 | 3 | 10 | 12 | 20 | 10 | 1.2 |
| 3 | 4 | 4 | 12 | 16 | 15 | 10 | 0.9 |
| 4 | 8 | 5 | 14 | 20 | 30 | 10 | 1.4 |
PS C:\Users\kevor\OneDrive\Desktop\Lucy_AI Training\hw 2\task 2>
```

```
PS C:\Users\kevor\OneDrive\Desktop\Lucy_AI Training\hw 2\task 2> & "C:\Program Files\Python312\python.exe
" "c:/Users/kevor/OneDrive/Desktop/Lucy_AI Training/hw 2/task 2/test.py"
run 1: enhancement = 13.9, image saved to focus_work\focus_run_1.png
PS C:\Users\kevor\OneDrive\Desktop\Lucy_AI Training\hw 2\task 2>
```